

- 11** A double star system is at a distance of 1.9×10^{17} m from the Earth. A telescope with a diameter of 3.0 m is used to view the stars.

What is the approximate minimum separation between the two stars of the double star system such that both stars can be detected by the telescope?

- A** 5.0×10^8 m **B** 1.0×10^9 m **C** 3.0×10^{10} m **D** 3.0×10^{11} m

11	C	$\theta \approx \frac{\lambda}{3} \approx \frac{x}{1.9 \times 10^{17}}$ If violet light: $x \approx \frac{400 \times 10^{-9}}{3} \times 1.9 \times 10^{17} = 2.5 \times 10^{10}$ m If red light: $x \approx \frac{700 \times 10^{-9}}{3} \times 1.9 \times 10^{17} = 4.4 \times 10^{10}$ m Since wavelength of visible light must be between wavelength of violet light and wavelength of red light, answer must be between 2.5 and 4.4×10^{10} m
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[Turn over

12 The diagram below represents equipotential lines of a gravitational field.